Question 1(d)

$$Prove (a+b)C = aC + bC$$

First, evaluate the sum of the scalars a and b.

$$a + b = 4 + (-7) = -3$$

Now, multiply the scalar result by the matrix C.

$$(a+b)C = -3 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} -12 & -3 \\ 9 & 6 \end{bmatrix}$$

Next, evaluate the right-hand side by calculating aC and bC separately.

$$aC = 4 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} 16 & 4 \\ -12 & -8 \end{bmatrix}$$

$$bC = -7 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} -28 & -7 \\ 21 & 14 \end{bmatrix}$$

Finally, add the results of aC and bC together.

$$aC + bC = \begin{bmatrix} 16 - 28 & 4 - 7 \\ -12 + 21 & -8 + 14 \end{bmatrix} = \begin{bmatrix} -12 & -3 \\ 9 & 6 \end{bmatrix}$$

Question 3(a)

$$(A^T)^T = A$$

First, find the transpose of matrix A by swapping its rows and columns.

$$A^T = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}$$

Now, transpose the result of A Transpose to return to the original orientation

$$(A^T)^T = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$$

Question 5

Calculate the Inverse of:

$$A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$$

Calculate the determinant of the matrix A.

$$\det(A) = (2)(4) - (-3)(4) = 8 + 12 = 20$$

Now, apply the 2×2 inverse formula by swapping the main diagonal, negating the off-diagonals, and dividing by the determinant

$$A^{-1} = \frac{1}{20}\begin{bmatrix} 4 & 3 \\ -4 & 2 \end{bmatrix}$$

Simplify the final matrix.

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{3}{20} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}$$

Question 7

$$C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Find the determinant of the diagonal matrix C.

$$\det(C) = (2)(3) - (0)(0) = 6$$

Apply the inverse formula.

$$C^{-1} = \frac{1}{6}\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Final simplified form:

$$C^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$$

Question 9

Find the inverse of:

$$\begin{bmatrix} \frac{1}{2}(e^{x} + e^{-x}) & \frac{1}{2}(e^{x} - e^{-x}) \\ \frac{1}{2}(e^{x} - e^{-x}) & \frac{1}{2}(e^{x} + e^{-x}) \end{bmatrix}$$

Calculate the determinant.

$$\det = \frac{1}{4}(e^{2x} + 2 + e^{-2x}) - \frac{1}{4}(e^{2x} - 2 + e^{-2x}) = 1$$

Because the determinant is 1, swap the main diagonal elements and negate the off-diagonal elements.

$$\begin{bmatrix} \frac{1}{2}(e^{x} + e^{-x}) & -\frac{1}{2}(e^{x} - e^{-x}) \\ -\frac{1}{2}(e^{x} - e^{-x}) & \frac{1}{2}(e^{x} + e^{-x}) \end{bmatrix}$$

Question 10

Find the inverse of the rotation matrix:

$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

Calculate the determinant using the Pythagorean identity:

$$cos^2\theta + sin^2\theta = 1.$$

$$\det = (\cos\theta)(\cos\theta) - (\sin\theta)(-\sin\theta) = \cos^2\theta + \sin^2\theta = 1$$

Swap the diagonal elements and negate the off-diagonal elements to find the inverse.

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Question 12

$$(A^{-1})^{-1} = A$$

Take the inverse of A found in Question 5.

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{3}{20} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}$$

Calculate the determinant of A Inverse.

$$\det(A^{-1}) = (\frac{1}{5})(\frac{1}{10}) - (\frac{3}{20})(-\frac{1}{5}) = \frac{1}{50} + \frac{3}{100} = \frac{5}{100} = \frac{1}{20}$$

Apply the inverse formula to A Inverse.

$$(A^{-1})^{-1} = 20 \begin{bmatrix} \frac{1}{10} & -\frac{3}{20} \\ \frac{1}{5} & \frac{1}{5} \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$$

Question 13

$$(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$$

This is a general property where the inverse of a product is the product of the inverses in reverse order.

$$\text{If } L = ABC, \text{ then } L^{-1} = C^{-1}B^{-1}A^{-1}$$

Question 16

Find a Given:

$$(5A^T)^{-1} = \begin{bmatrix} -3 & -1 \\ 5 & 2 \end{bmatrix}$$

Invert both sides of the equation to isolate 5A Transpose.

$$5A^T = \left(\begin{bmatrix} -3 & -1 \\ 5 & 2 \end{bmatrix}\right)^{-1}$$

Calculate the inverse. The determinant is (−3)(2)−(−1)(5)=−1.

$$5A^T = \frac{1}{-1}\begin{bmatrix} 2 & 1 \\ -5 & -3 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 5 & 3 \end{bmatrix}$$

Divide by 5 and then transpose to find A.

$$A^T = \begin{bmatrix} (\frac{-2}{5}) & (\frac{-1}{5}) \\ (\frac{5}{5}) & (\frac{3}{5}) \end{bmatrix} \implies A = \begin{bmatrix} (\frac{-2}{5}) & 1 \\ (\frac{-1}{5}) & (\frac{3}{5}) \end{bmatrix}$$

Question 17

$$(I + 2A)^{-1} = \begin{bmatrix} -1 & 2 \\ 4 & 5 \end{bmatrix}$$

Invert both sides.

$$I + 2A = \left(\begin{bmatrix} -1 & 2 \\ 4 & 5 \end{bmatrix}\right)^{-1}$$

The determinant is (−1)(5)−(2)(4)=−13.

$$I + 2A = \frac{1}{-13}\begin{bmatrix} 5 & -2 \\ -4 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{5}{13} & \frac{2}{13} \\ \frac{4}{13} & \frac{1}{13} \end{bmatrix}$$

Subtract I and divide by 2.

$$2A = \begin{bmatrix} -\frac{18}{13} & \frac{2}{13} \\ \frac{4}{13} & -\frac{12}{13} \end{bmatrix} \implies A = \begin{bmatrix} -\frac{9}{13} & \frac{1}{13} \\ \frac{2}{13} & -\frac{6}{13} \end{bmatrix}$$

Question 19(a)

$$A^3 \text{ for } A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

First, find:

$$A^2$$

$$A^2 = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix}$$

Now, multiply A Square by A

$$A^3 = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 41 & 15 \\ 30 & 11 \end{bmatrix}$$

Question 20(a)

Question: Compute A^3 using the given matrix $A = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$.

First, compute A^2 by multiplying matrix A by itself.

$$A^2 = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} (2)(2) + (0)(4) & (2)(0) + (0)(1) \\ (4)(2) + (1)(4) & (4)(0) + (1)(1) \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 12 & 1 \end{bmatrix}$$

Next, compute A^3 by multiplying A^2 by A.

$$A^3 = \begin{bmatrix} 4 & 0 \\ 12 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} (4)(2) + (0)(4) & (4)(0) + (0)(1) \\ (12)(2) + (1)(4) & (12)(0) + (1)(1) \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 28 & 1 \end{bmatrix}$$

Question 21

Compute $p(A)$ for the matrix $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ and the following polynomials:

(a) $p(x) = x - 2$,
(b) $p(x) = 2x^2 - x + 1$, and
(c) $p(x) = x^3 - 2x + 1$.

Part (a): For the polynomial $p(x) = x - 2$, substitute x with matrix A and the constant 2 with $2I$, where I is the 2×2 identity matrix.

$$p(A) = A - 2I = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$$

Part (b): For $p(x) = 2x^2 - x + 1$, we first need to find A^2.

$$A^2 = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix}$$

Now, substitute A and I into the polynomial $p(A) = 2A^2 - A + I$.

$$p(A) = 2 \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiply by the scalar and combine the matrices.

$$p(A) = \begin{bmatrix} 22 & 8 \\ 16 & 6 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 20 & 7 \\ 14 & 6 \end{bmatrix}$$

Part (c): For $p(x) = x^3 - 2x + 1$, we first need to find A^3 by multiplying A^2 by A.

$$A^3 = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 41 & 15 \\ 30 & 11 \end{bmatrix}$$

Substitute the matrices into $p(A) = A^3 - 2A + I$.

$$p(A) = \begin{bmatrix} 41 & 15 \\ 30 & 11 \end{bmatrix} - 2 \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Perform the scalar multiplication and then the matrix addition and subtraction.

$$p(A) = \begin{bmatrix} 41 & 15 \\ 30 & 11 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 36 & 13 \\ 26 & 10 \end{bmatrix}$$

Question 22

Compute $p(A)$ for the matrix $A = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$ and the same polynomials as Question 21.

First, note that we have already calculated A^2 and A^3 for this matrix in Question 20(a).

$$A = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}, \quad A^2 = \begin{bmatrix} 4 & 0 \\ 12 & 1 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 8 & 0 \\ 28 & 1 \end{bmatrix}$$

Part (a): For $p(x) = x - 2$, evaluate $p(A) = A - 2I$.

$$p(A) = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 4 & -1 \end{bmatrix}$$

Part (b): For $p(x) = 2x^2 - x + 1$, evaluate $p(A) = 2A^2 - A + I$.

$$p(A) = 2 \begin{bmatrix} 4 & 0 \\ 12 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 8 & 0 \\ 24 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 20 & 2 \end{bmatrix}$$

Part (c): For $p(x) = x^3 - 2x + 1$, evaluate $p(A) = A^3 - 2A + I$.

$$p(A) = \begin{bmatrix} 8 & 0 \\ 28 & 1 \end{bmatrix} - 2 \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 8 & 0 \\ 28 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 8 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 20 & 0 \end{bmatrix}$$

Question 37

Question: Determine whether $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ is invertible, and if so, find the inverse.

To find the inverse, we can use the Gauss-Jordan elimination method by creating an augmented matrix $[A|I]$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Subtract Row 1 from Row 2 ($R_2 \leftarrow R_2 - R_1$).

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Subtract Row 2 from Row 3 ($R_3 \leftarrow R_3 - R_2$).

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 \end{array}\right]$$

Divide Row 3 by 2 ($R_3 \leftarrow \frac{1}{2}R_3$).

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right]$$

Add Row 3 to Row 2 ($R_2 \leftarrow R_2 + R_3$).

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right]$$

Subtract Row 3 from Row 1 ($R_1 \leftarrow R_1 - R_3$).

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right]$$

Since the left side has been transformed into the identity matrix, matrix A is invertible. The right side is the inverse matrix A^{-1}.

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Question 38

Determine whether $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ is invertible, and if so, find the inverse.

We set up the augmented matrix $[A|I]$ to test for invertibility.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Swap Row 1 and Row 2 ($R_1 \leftrightarrow R_2$) to simplify the pivot matrix.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Subtract Row 1 from Row 2 ($R_2 \leftarrow R_2 - R_1$).

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Subtract Row 2 from Row 3 ($R_3 \leftarrow R_3 - R_2$).

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{array}\right]$$

Since the entire third row of the left side is composed of zeros, it is impossible to transform it